

COURSE OUTLINE



Basic Information:		
Course Code: CSE 411		
Course Title: Computer Graphics		
Credits: 3	Weekly: 3 hours	Total Week: 14

Prerequisite:

MAT211, CSE213, CSE222, STA 223

Course Content (from syllabus):

Introduction: A Survey of Computer Graphics, Application of Computer Graphics, Video Display Devices: Refresh Cathode Ray Tubes, Raster and Random Scan Displays, Color CRT Monitors, DVST, Flat-Panel Displays, Points and Lines, Line Drawing Algorithm, DDA Algorithm, DDA Algorithm example with plot in a graph, Bresenham's Line Algorithm with parameter description, Bresenham's Line Algorithm with example, Circle Generating Algorithm, Properties of Circle, Midpoint Circle Algorithm, Midpoint Circle Algorithm example, 2D Geometric Transformation with example, 3D Transformation with example, Antialiasing and different techniques, Color Models and Color Applications, Two-Dimensional Viewing, Window-Viewport Mapping, Clipping, Cohen-Sutherland Line Clipping Algorithm, Cohen-Sutherland Line Clipping Algorithm example, Polygon Clipping using Sutherland-Hodgeman Algorithm, Sutherland-Hodgeman Polygon Clipping example, Polygon Clipping using Weiler-Atherton Algorithm, Weiler-Atherton Polygon Clipping example, Three-Dimensional Display Methods, Parallel Projection, Perspective Projection, Numerical Problems on Projections, Depth Cueing, Visible Line and Surface Identification, Bezier Curves, Spline Curves, Bezier and Spline Curves examples, Halftone Pattern, Dithering Techniques, Computer Animation and Design of Animation Sequences, Multimedia Systems, Data Compression, Huffman Coding, Lossy Compression Methods: JPEG and MPEG.

Course Description/ Rationale:

Computer Graphics is a 3-credit senior-level course that introduces the concepts and implementation of computer graphics. As one of the important subject areas of the study of computer science and information systems, this course will focus on the theoretical aspects and implementation of computer graphics using OpenGL.

Computer Graphics and Design - Foundation provides potential for the engagement of integrated learning opportunities and the capacity to develop design thinking skills to effectively transfer knowledge and understanding across disciplines. The applied design thinking and problem-solving focus of this course helps equip learners to develop skills essential for the digital age.

Course Objective:

- To introduce the concepts of computer graphics.
- To gather knowledge about 2D graphics with lines and curves and implement algorithms to rasterize simple shapes, fill and clip polygons, and have a basic grasp of anti-aliasing techniques.
- To learn the geometrical transformations and 2D viewing.
- To learn the techniques for representing 2D and 3D geometrical objects.

- To learn the color models, lighting and shading models, animation, dithering, hidden surface elimination, and rendering.

Course Outcomes (COs) with Mappings:

After completion of this course, students will be able to:

COs	CO Statements	POs	Learning Domains	Knowledge Profile	Complex Engineering Problem	Complex Engineering Activities
CO1	Explain the core concepts of computer graphics, including primitives, color models, lighting, shading, animation, hidden surface elimination, and rendering.	PO1	C1, C2	K2-K4	EP1	
CO2	Analyze computer graphics algorithms such as line drawing, transformations, and clipping to solve graphical problems.	PO2	C2, C3, C4	K2-K4	EP1, EP2	
CO3	Apply computer graphics algorithms to investigate their computational behavior and limitations	PO4	C1, C2, C3	K8	EP1	

Bloom's Taxonomy

Cognitive Domain

C1: Remember

C2: Understand

C3: Apply

C4: Analyze

C5: Evaluate

C6: Create

Psychomotor Domain

P1: Imitation

P2: Manipulating

P3: Precising

P4: Articulating

P5: Naturalization

Affective Domain

A1: Receiving/ Attending

A2: Responding

A3: Valuing,

A4: Organizing

A5: Characterizing

Knowledge Profile

K1: Fundamental Knowledge

K2: Mathematics

K3: Engineering Fundamentals

K4: Specialist Knowledge

K5: Engineering Design

K6: Technology

K7: Society

K8: Research

CEP Attributes

EP1: Depth of Knowledge

EP2: Range of Conflicting

Requirements

EP3: Depth of analysis

required.

EP4: Familiarity of issues

EP5: Extend of Applicable Code

EP6: Extend of Stakeholders

involvements and Conflicting

Requirements

EP7: Interdependence

CEA Attributes

EA1: Range of resources

EA2: Level of interaction

EA3: Innovation

EA4: Consequences for society and

the environment

EA5: Familiarity

Justification of CO-PO Mapping:

COs	POs	Justification of Mapping
CO1	PO1	CO1 enables students to understand the fundamental concepts of computer graphics, including primitives, color models, lighting, shading, animation, hidden surface removal, and rendering, thereby strengthening their foundational engineering knowledge.
CO2	PO2	CO2 requires students to analyze graphics algorithms such as output primitives and 2D/3D transformations, supporting their ability to formulate and solve graphical problems using appropriate mathematical and algorithmic approaches
CO3	PO4	CO3 requires students to apply computer graphics algorithms to investigate and analyze their computational behavior and limitations, thereby supporting the investigation of complex graphical problems using analytical and research-based approaches.

Justification of Knowledge Profile (KP) Mapping:

COs	Knowledge Profile (K)	Justification of Mapping
CO1	K2, K3, K4	CO1 focuses on explaining fundamental computer graphics concepts such as primitives, color models, lighting, shading, animation, hidden surface elimination, and rendering. This aligns with K2 for understanding core engineering fundamentals, K3 for acquiring specialized graphics knowledge, and K4 for interpreting and analyzing graphical methods and techniques.
CO2	K2, K3, K4	CO2 involves analyzing computer graphics algorithms including line drawing, transformations, and clipping. This supports K2 by reinforcing fundamental principles, K3 through the application of specialized graphics knowledge, and K4 by enabling analytical comparison and use of algorithmic methods to solve graphical problems
CO3	K8	CO3 emphasizes applying graphics algorithms to investigate and analyze their computational behavior and limitations. This aligns with K8 as it requires research-oriented analytical thinking and investigation of algorithmic performance using theoretical and methodical approaches.

Justification of Complex Engineering Problem (CEP) Attributes Mapping:

COs	CEP Attributes (EP)	Justification of Mapping
CO1	EP1	CO1 requires understanding fundamental concepts of computer graphics such as primitives, color models, lighting, shading, animation, hidden surface elimination, and rendering. These concepts demand a strong depth of engineering knowledge and conceptual understanding, which aligns with EP1 (Depth of Knowledge).
CO2	EP1, EP2	CO2 involves analyzing computer graphics algorithms, including line drawing, transformations, and clipping. Solving these problems requires in-depth engineering knowledge (EP1) and addressing different technical requirements and trade-offs among algorithms, which corresponds to EP2 (Range of Conflicting Requirements).
CO3	EP1	CO3 focuses on applying graphics algorithms to investigate their computational behavior and limitations. This requires solid foundational knowledge and analytical reasoning about algorithm performance, aligning with EP1 (Depth of Knowledge).

Justification of Complex Engineering Activities (CEA) Attributes Mapping (if any):

COs	CEA Attributes (EA)	Justification of Mapping
CO1	N/A	N/A
CO2	N/A	N/A
CO3	N/A	N/A

Course Delivery Plan/Lesson Delivery Plan:

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies
1	Lesson 1: Introduction: A Survey of Computer Graphics, Application of Computer Graphics (Text, 4-34)	Lesson 1: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1	PO1	Midterm
	Lesson 2: Video Display Devices: Refresh Cathode Ray Tubes, Raster and Random Scan Displays, Color CRT Monitors, DVST, Flat-Panel Displays. (Text, 36-52)	Lesson 2: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1	PO1	
2	Lesson 3: Points and Lines, Line Drawing Algorithm, DDA Algorithm (Text. 84-88)	Lesson 3: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1, CO2, CO3	PO1, PO2, PO4	Class Test, Midterm
	Lesson 4: DDA Algorithm-example with plot in a graph.	Lesson 4: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
3	Lesson 5: Bresenham's Line Algorithm with Parameter description. (Text, 88-92)	Lesson 5: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1, CO2, CO3	PO1, PO2, PO4	Class Test, Midterm
	Lesson 6: Bresenham's Line Algorithm with example	Lesson 6: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies
4	Lesson 7: Circle Generating Algorithm, Properties of Circle, Midpoint Circle Algorithm (Text, 97-101)	Lesson 7: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1, CO2	PO1, PO2	Assignment, Midterm
	Lesson 8: Midpoint Circle Algorithm with example	Lesson 8: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
5	Lesson 9: 2D Geometric Transformation with example	Lesson 9: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1, CO2	PO1, PO2	Class Test, Midterm
	Lesson 10: 3D Transformation and example	Lesson 10: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
6	Lesson11: Antialiasing-Different techniques (Text, 171-178)	Lesson 11: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1	PO1	Midterm
	Lesson 12: Color Models and Color Applications (Text, Chap-15, 572-575)	Lesson 12: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
7	Midterm Exam				
8	Lesson 13: Two-Dimensional Viewing, Window-Viewport mapping, Clipping, Cohen-Sutherland Line Clipping Algorithm, Example	Lesson 13: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1. CO2	PO1, PO2	Assignment, Final Exam
	Lesson 14: Polygon Clipping: Sutherland-Hodgeman Polygon Clipping Algorithm (Text, 237-242)	Lesson 14: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT,	CO2, CO3	PO2, PO4	Class Test, Assignment, Final Exam

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies
		Video, H5P;			
9	Lesson 15: Sutherland-Hodgeman Polygon Clipping Algorithm-Example	Lesson 15: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO2, CO3	PO2, PO4	Class Test, Assignment, Final Exam
	Lesson 16: Polygon Clipping: Weiler-Atherton Polygon Clipping Algorithm, Examples (Text, 242-243)	Lesson 16: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
10	Lesson 17: Three Dimensional Display Methods, Parallel Projection, Perspective projection, Numerical Problems on Projections. (Text, Chap-9, 297-299)	Lesson 17: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1, CO2	PO1, PO2	Final Exam
	Lesson 18: Depth Cueing and Visible Line/Surface Identification (Text, Chapter 9, Pages 350–365)	Lesson 18: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1, CO2	PO1, PO2	
11	Lesson 19: Bezier Curves, Spline Curves, Examples (Text, Chapter-10, 327-346)	Lesson 19: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1	PO1	Final Exam
	Lesson 20: Halftone Pattern and Dithering Techniques (Text, Chapter-14, 516-522)	Lesson 20: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
12	Lesson 21: Computer Animation: Design of Animation Sequences (Text, Chap-16, 299-301)	Lesson 21: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1	PO1	Final Exam
	Lesson 22: Multimedia system, Data compression, Huffman coding	Lesson 22: Online/Onsite discussion; Review			

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies
		Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			
13	Lesson 23: Lossy Compression Methods: JPEG, MPEG	Lesson 23: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;	CO1	PO1	Final Exam
	Lesson 24 Review class for final exam	Lesson 24: Online/Onsite discussion; Review Feedback Online; Using interactive content e.g. Voice over PPT, PPT, Video, H5P;			Final Exam (Either online or onsite based on Wk9-Wk16 discussion)
14	Final Exam				

PO Validation via Assessment:

Assessment Type	Mapped COs	Mapped POs	Justification
Midterm	CO1, CO2	PO1, PO2, PO4	The Midterm examination assesses students' understanding of fundamental computer graphics concepts such as display devices, color models, and rendering (CO1), their ability to analyze and apply line drawing and transformation algorithms (CO2), and their capability to investigate the computational behavior and limitations of basic graphics algorithms (CO3). This supports the attainment of foundational engineering knowledge (PO1), analytical problem-solving skills (PO2), and investigative competence in graphical algorithms (PO4).
Final	CO1, CO2, CO3	PO1, PO2, PO4	The Final examination evaluates comprehensive knowledge of advanced computer graphics topics including viewing transformations, clipping algorithms, projections, curves, animation, rendering, and

			compression techniques (CO1), analysis of graphics algorithms and transformations (CO2), and application of graphics algorithms to investigate their computational behavior and limitations in graphical applications (CO3). This ensures achievement of core engineering knowledge (PO1), analytical ability (PO2), and investigation of complex graphical problems (PO4).
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Assessment Pattern:

Assessment Tools	CO-wise Marks				Marks (Total=100)
	CO1	CO2	CO3	CO4	
Attendance	Not Applicable				
Quiz	Not Applicable				
Presentation	Not Applicable				
Assignment	Not Applicable				
Mid-Term	10	15			25
Final-Term	15	15	10		40
Total	25	30	10		65

CIE – Breakup (Theory) [60 marks]

Bloom's Criteria	Attendance	Class Test	Assignment	Presentation	Mid Exam	Total
	07	15	05	08	25	60
Remember						
Understand						
Apply						
Analyze						
Evaluate						
Create						

SEE – Semester End Examination [40 marks]

Bloom Criteria	Score for the Test
Remember	
Understand	10
Apply	10
Analyze	20

Evaluate	
Create	

Learning Materials:

Textbook/Recommended Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1.	Computer Graphics	Donald Hearn, M. Pauline Baker	Prentice Hall / Pearson	2 nd Edition

Reference Books/Supplementary Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1	Schaum's Outline of Computer Graphics	Ray Plastock, Gordon Kalley, Zhiang Xiang, Zhingang Xiang	McGraw-Hill	2nd Edition, 2000
2	Fundamentals of Computer Graphics	Peter Shirley	CRC Press (Taylor & Francis Group)	4th Edition, 2015

Prepared by Module Leader

Name _____

Signature _____

Approved by OBE Committee
